# Supplementary Material

### A. Computational Resources and Efficiency Tradeoffs

Compared with existing models, SkeDiff achieves high-quality 3D reconstruction using only two orthogonal X-ray images and a latent space diffusion architecture, whereas other methods often rely on multiple views or more complex designs. For example, models such as DCDS [1], DiffuX2CT [2], and X-CTRSNet [3] utilize multi-view inputs and larger network structures, which typically require comparable or higher computational resources. However, despite the similar or greater resource consumption, their reconstruction performance on sparse-view or limited-input scenarios is generally lower than that of SkeDiff. In terms of inference efficiency, SkeDiff takes approximately 900 ms per case for a 128×128×128 volume, with GPU memory usage of about 12 GB. While we do not claim SkeDiff is faster than existing approaches overall, it achieves a favorable balance between computational cost and reconstruction quality. Benefiting from its latent space diffusion framework, SkeDiff is also notably more efficient than pixel-space diffusion models, which often incur substantially higher costs due to high-dimensional denoising operations. Overall, SkeDiff incurs moderate computational load while producing detailed and clinically relevant skeletal reconstructions. In future work, we plan to explore fast sampling strategies to further reduce inference time without sacrificing accuracy.

### B. Robustness to Image Quality Degradation

To further evaluate the robustness of SkeDiff, we conducted experiments introducing Gaussian noise, scatter simulation, and low-dose X-ray conditions, which are common challenges in clinical imaging. As illustrated in Fig. **??**, the first row shows the ground truth CT images, the second row presents reconstruction results from SkeDiff using clean input X-rays, and the third row shows results when the input X-rays were degraded with Gaussian noise and scatter simulation. While the reconstruction quality in the noisy scenario is somewhat reduced, the bone structures and anatomical details remain largely preserved. This indicates that SkeDiff is able to maintain reasonable accuracy in 3D reconstructions even under image quality degradation. These findings suggest that SkeDiff holds potential for use in clinical settings where input X-rays may be affected by noise or acquired under low-dose protocols.

### C. Training Stability and Practical Considerations

We evaluated the impact of batch size on model performance by conducting experiments with different global batch sizes (2×4, 2×6, 2×8) using four, six and eight GPUs respectively, while keeping all other hyperparameters constant. As detailed in Table I, the quantitative metrics remained consistent. We additionally provide the training loss and validation metric curves over 200 epochs in Fig. 1 to demonstrate stable convergence.

TABLE I: Hyperparameter sensitivity analysis for batch size. GBZ represents the global batch size. NG represents the number of GPU. The experiments were conducted on spine dataset.

| GBZ | NG | PSNR ↑ | SSIM ↑ | CS ↑ | MAE ↓ | LPIPS ↓ |
|---|---|---|---|---|---|---|
| 8 | 4 | 33.4811 | 0.8066 | 0.8012 | 66.9049 | 0.4959 |
| 12 | 6 | 33.0458 | 0.8005 | 0.7949 | 66.4521 | 0.4924 |
| 16 | 8 | 33.7155 | 0.8122 | 0.8075 | 67.3508 | 0.5093 |

TABLE II: Variance across random seeds on spine dataset.

| Seed | PSNR ↑ | SSIM ↑ | CS ↑ | MAE ↓ | LPIPS ↓ |
|---|---|---|---|---|---|
| 0 | 33.2474 | 0.8047 | 0.8139 | 67.2679 | 0.4971 |
| 1 | 33.4348 | 0.8189 | 0.8073 | 66.5949 | 0.5164 |
| 2 | 33.1980 | 0.7987 | 0.7974 | 66.5933 | 0.5073 |
| Mean | 33.29349 | 0.8074 | 0.8062 | 66.8187 | 0.5069 |
| ±Std | ±0.1249 | ±0.0104 | ±0.0083 | ±0.3890 | ±0.0097 |

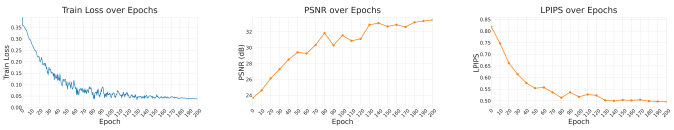


Fig. 1: Visualization of the training loss (left), PSNR (middle), and LPIPS (right) over 200 epochs on the spine dataset.

We evaluated the robustness of SkeDiff by conducting three independent training runs with different random seeds. The results in Table II show a standard deviation of approximately 1%. The minimal variance across runs highlights the high reproducibility of our approach. To save computational resources, the experiments were repeated using the spine as a representative dataset.

Regarding the optimizer, we selected SGD over adaptive methods like AdamW. Our empirical experiments indicated that SGD provided superior generalization on the test set, likely by avoiding the sharp minima often associated with adaptive optimizers in low-data medical imaging regimes. The momentum in SGD, coupled with our effective global batch size of 8, ensured efficient and stable.

## References

[1] Z. Li, D. Chang, Z. Zhang, F. Luo, Q. Liu, J. Zhang, G. Yang, and W. Wu, "Dual-domain collaborative diffusion sampling for multi-source stationary computed tomography reconstruction," *IEEE Transactions on Medical Imaging*, 2024.

[2] X. Liu, Z. Qiao, R. Liu, H. Li, J. Zhang, X. Zhen, Z. Qian, and B. Zhang, "Diffux2ct: Diffusion learning to reconstruct ct images from biplanar x-rays," in *European Conference on Computer Vision*. Springer, 2024, pp. 458–476.

[3] R. Ge, Y. He, C. Xia, C. Xu, W. Sun, G. Yang, J. Li, Z. Wang, H. Yu, D. Zhang *et al.*, "X-ctrsnet: 3d cervical vertebra ct reconstruction and segmentation directly from 2d x-ray images," *Knowledge-Based Systems*, vol. 236, p. 107680, 2022.